

Smart Attendance Using RFID

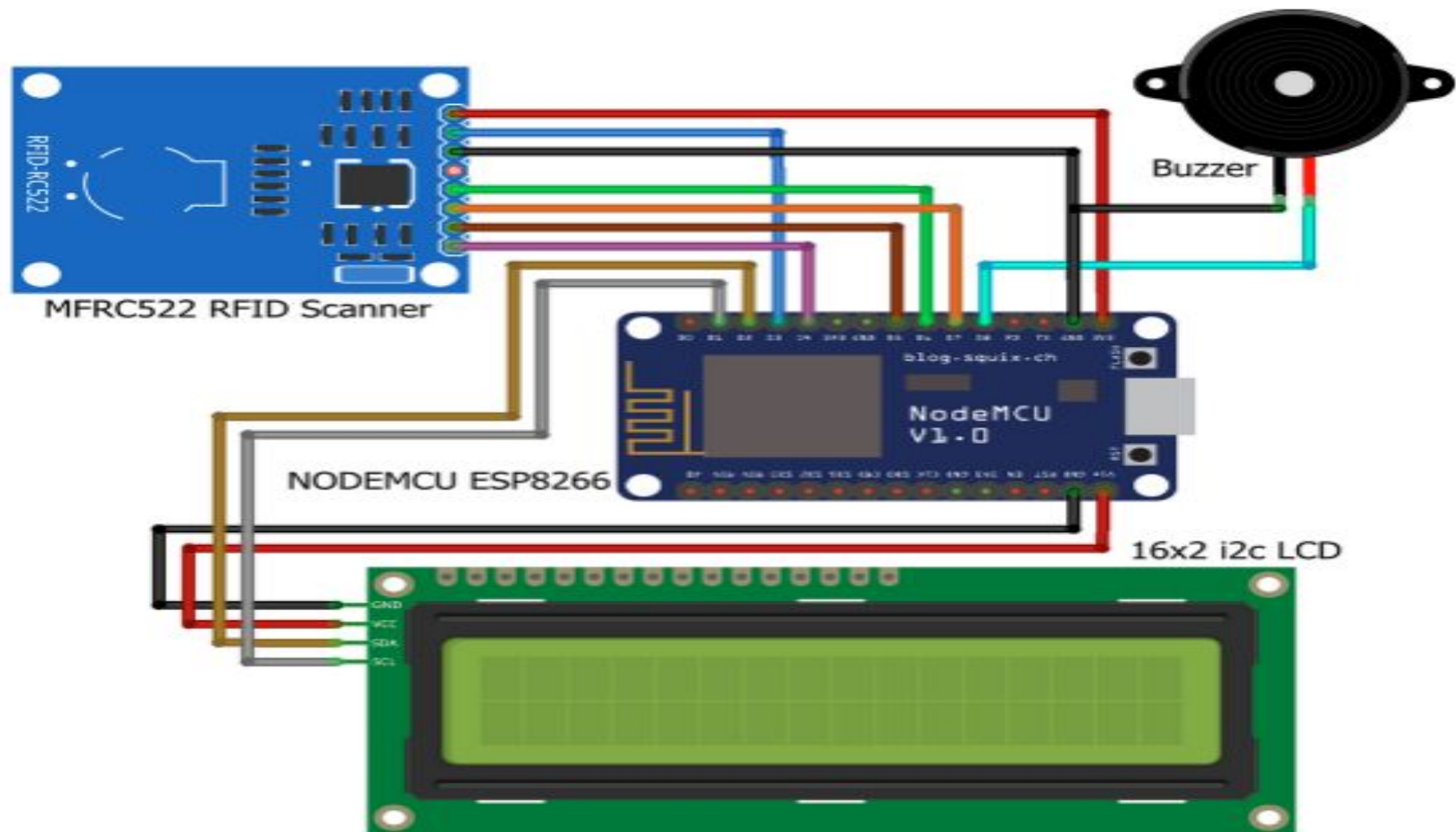
Wireless Technology

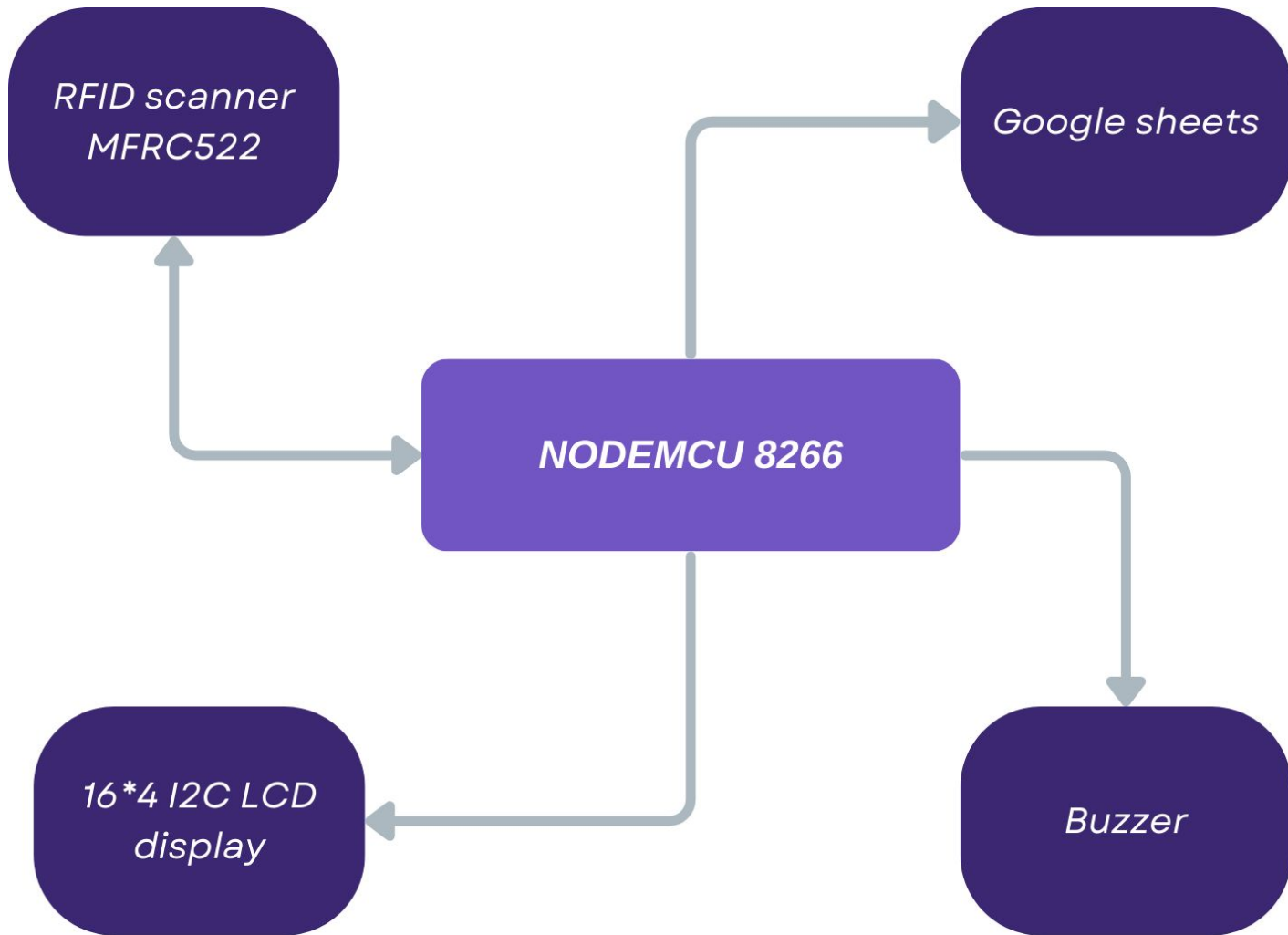
Introduction

This system leverages **Radio-Frequency Identification (RFID)** technology to enable quick and contactless identification of individuals. Each participant is issued a unique RFID tag, which, upon scanning, communicates with the NodeMCU ESP8266. The microcontroller processes the tag information and transmits it over Wi-Fi to update attendance records in a Google Sheets document instantaneously.

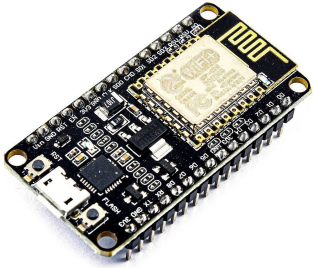
Key Features:

- **Real-Time Data Logging:** Attendance records are updated in Google Sheets immediately after scanning, ensuring up-to-date information.
- **Remote Accessibility:** Authorized personnel can access and monitor attendance data from any location via Google Sheets.
- **Efficiency and Accuracy:** Automated data entry reduces human error and streamlines the attendance-taking process.



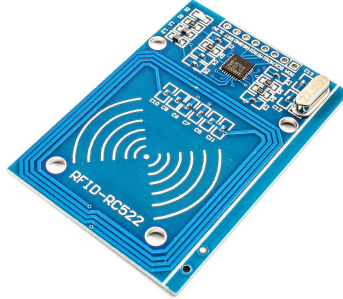


Components required



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Nodemcu esp8266

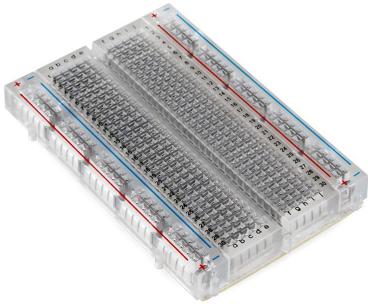


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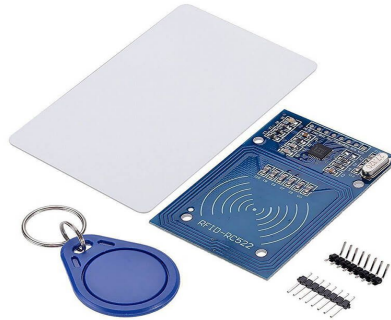
RFID reader



LCD 16*4 display



breadboard



RFID tags



buzzer

RFID Sensor Specification

- Frequency - 13.56MHz
- Supply voltage - 3.3V
- Range - 2-5 cm
- RFID tag data storage - 1kb

Sensors

Definition: RFID stands for Radio Frequency Identification, a technology that uses radio waves to identify and track objects, animals, or people.

Functionality: It allows for non-contact data transfer, making it useful in various industries where traditional barcode systems may not be suitable

Working Principle:

- RFID reader emits radio waves.
- RFID tag responds by transmitting stored data.
- Reader processes the data for identification.
- Data Reception and Processing
- Data Analysis and Application

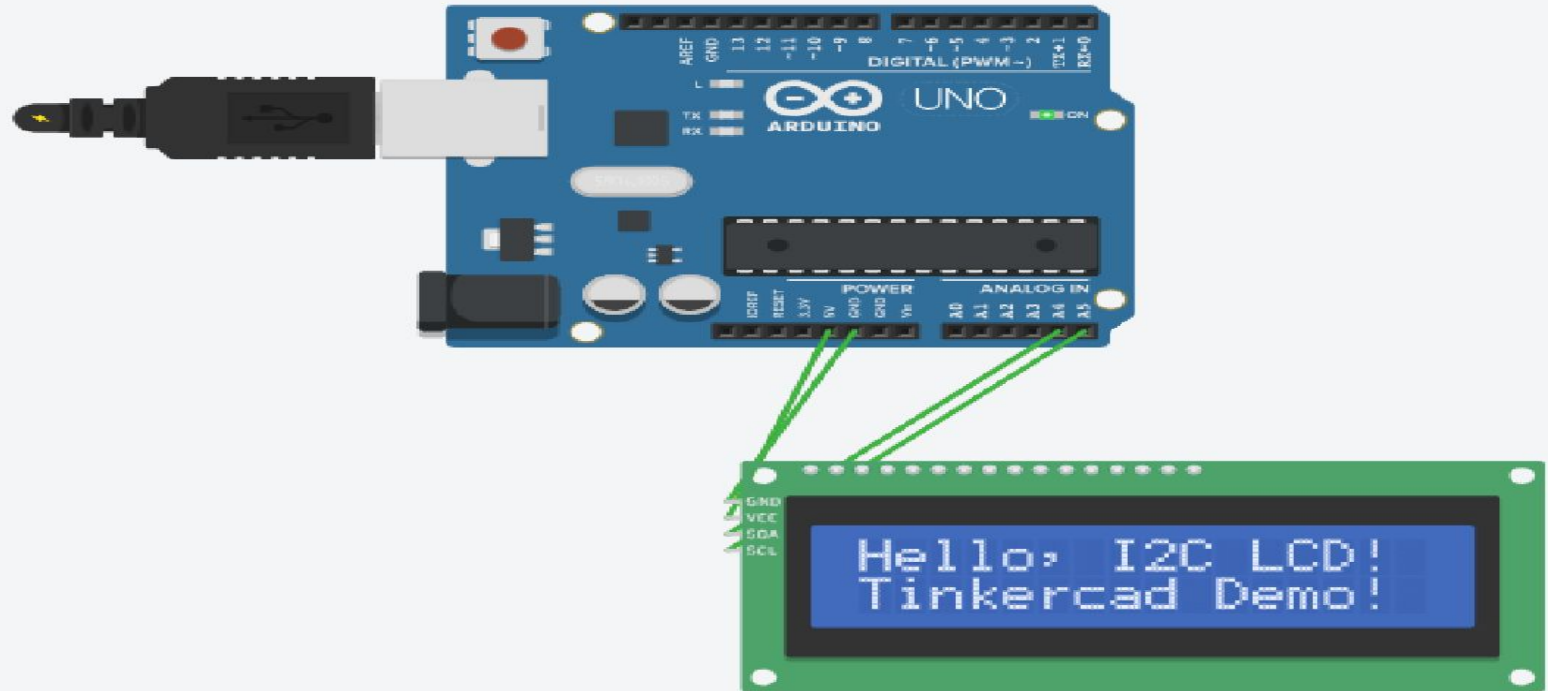
Sensors

Components of an RFID System

An RFID system typically consists of the following components:

1. RFID Tags (Transponders)
 - Integrated Circuit (IC): Stores unique identification data and manages communication with the reader.
 - Antenna: Enables communication between the tag and the reader. In passive tags, it also captures energy from the reader's signal to power the IC.
 - Substrate and Packaging: The tag's components are mounted on a substrate and enclosed in protective packaging.
2. RFID Readers (Interrogators)
 - Antenna: Generates an electromagnetic field to activate passive tags and receives data from them.
 - Transceiver: Sends and receives radio frequency signals to and from the tags.
 - Controller: Processes the received data and communicates with external systems.
3. RFID Software
 - Software Applications: Manage and analyze the data collected by the RFID system.

Simulation on tinkercad



Connections

NodeMcu esp8266 to RFID reader

VCC	3.3V
GND	GND
SDA	D2
SCK	D5
MOSI	D7
MISO	D6
RST	D4
VCC	3.3V

Connections

NodeMCU 8266 to LCD pin

VCC	3.3V
GND	GND
SDA	D1
SCL	D2

NodeMCU to buzzer pin

Positive	D8
Negative	GND

Future scope

- Currently our project is perfectly giving results for the in-time after scanning the rfid tag, in the future we might also make it for out-time.
- The in-time information is currently stored in Google sheets, but in the future we might integrate it in real time, and at a larger scale.
- In the future we might also develop a hardware component, so that users can scan their tags and mark their attendance.